Daksh Adhar

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EDUCATION

Carnegie Mellon University - School of Computer Science Pittsburgh, PA
Master of Science in Robotic Systems Development (MRSD) | GPA: 4.05/4 May 2026

Indian Institute of Technology, Guwahati Guwahati, India
Bachelor of Technology in Engineering Physics | GPA: 8.27/10.00 May 2024

Courses: Deep RL, Deep Learning, Learning for 3D Vision, Computer Vision, Optimal Control, Robot Mobility, Autonomy

SKILLS

Python, C/C++, MATLAB, Julia, PyTorch, Tensorflow, MuJoCo, Isaac Gym, PyBullet, OpenAI Gym, ROS2, MoveIt2, Gazebo

EXPERIENCE

1X Technologies Palo Alto, CA
AI Resident, Reinforcement Learning Team May 2025 – August 2025

- Trained RL policies for dexterous manipulation on NEO hands, and added randomization for **sim-to-real** transfer
- Designed metrics to benchmark policies on **Isaac Gym** and **MuJoCo**, evaluating **sim-to-sim** robustness
- Developed a **ROS 2 Humble** C++ controller to deploy evaluated policies in real-time simulation and teleoperation
- Built Tkinter-based local and Streamlit-based browser app for object segmentation mask data collection using **SAM2**
- Integrated Cloudflare R2 and DBeaver **SQL** backend to load frames and store operator clicks on 1M+ frames

MaxLab CMU Pittsburgh, PA
Research Assistant, Prof. Max Simchowitz January 2026 – Present

- Built a **teleoperation** system (100Hz) for a dexterous hand on **UR5e** using Leap Motion for tracking and IK for control
- Developing **neural hand retargeting** by training a personalized mapping from human hand data to robot joint angles
- Extending a benchmark suite (**PushT**) to evaluate generative policies across task generalizability and difficulty axes

Biomimetic Robotics & Artificial Intelligence Laboratory Guwahati, India
Research Assistant, Prof. Shyamanta M. Hazarika January 2023 – May 2024

- Built a testing framework in **PyBullet** and **OpenAI Gym** to train RL policies for bionic hand power grasping
- Formulated reward functions and used **Soft Actor-Critic** algorithm to enable grasp-and-lift of deformable objects
- Applied **domain randomization** for sim-to-sim, achieving 38% slip reduction and 14% decrease in deformation

PROJECTS

Hierarchical VLM for Vision-Language Navigation on mobile robots | Code | Report September 2025 – October 2025

- Built an Input Handler to parse and route natural **language queries** for numerical, object reference, and navigation tasks
- Implemented **A*** exploration with loop-minimizing heuristics, achieving 82% map generation success across 17 scenes
- Enabled **spatial reasoning** via 3D-to-2D projection for object color association and geometric predicate prompting

Combining Specialist Policies for Generalist Multi-Agent Play | Code | Report September 2025 – December 2025

- Engineered a **Mixture-of-Experts** using a gating network to compose specialist policies from shared latent embeddings
- Implemented distributed **self-play** using **IMPALA** with **PPO** to stabilize learning against non-stationary opponents
- Outperformed specialist and generalist baselines achieving 2nd place by gating policies to improve gameplay

AR assisted Robotic Total Knee Arthroplasty | Website October 2024 – December 2025

- Developed a surgical robot for total knee replacement, achieving **2 mm and 2°** accuracy in drilling surgical pins
- Utilized **DINOv3** for bone cluster detection, **SAM** for segmentation, and **ICP** for real-time surgical plan registration
- Designed a **ROS2/MoveIt2** planning subsystem for pose-based drilling and collision aware trajectory planning
- Implemented an **Apple Vision Pro** AR interface for real-time drill site visualization and complete system control

Unitree G1 Humanoid Soccer Ball Kicking | Code | Report January 2025 – April 2025

- Simulated penalty-style kicking on Unitree G1 using **QP** for strike impulse and **DIRCOL** for trajectory optimization
- Stabilized whole-body motion with **IHLQR** and tracked kicking with **TVLQR**, achieving <0.5m error from targets

PUBLICATIONS

CMU Vision-Language-Autonomy Challenge (Presentation) October 2025
IROS 2025; AI Meets Autonomy: Vision Language and Autonomous Systems Workshop IROS Workshop

Grasp force optimization as a Bilinear Matrix inequality problem: A Deep-learning approach December 2023
6th National Conference on Multidisciplinary Design, Analysis and Optimization arXiv/2312.05034

Reinforcement Learning-Based Bionic Reflex Control for Anthropomorphic Robotic Grasping November 2023
Journal Paper at Cambridge University Press, Robotica Cambridge/Robotica